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Understanding Hip Replacement

A patient's guide to conventional, anterior-approach, robotic-assisted and day-care (ERAS) hip replacement surgery — everything worth knowing before your consultation.

4 Approaches

Explained side-by-side

Week-by-week

Recovery roadmap

95%+

Reported pain relief



Dr. Harshil Shah

M.S. Orthopaedics · Joint Replacement, Arthroscopy & Trauma

BEFORE YOU BEGIN

What this guide covers

A complete walk-through — from recognising the symptoms, to choosing an approach, to the physiotherapy that carries you home. Written for education, not diagnosis: use it to prepare sharper questions for your own consultation.

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| 01 Do you need a hip replacement? | 05 Implants & robotic assistance |
| 02 Basic anatomy of the hip | 06 Day-care (ERAS) hip replacement |
| 03 Types of hip replacement surgery | 07 Your journey — pre-op to post-op |
| 04 Anterior vs. posterior approach | 08 Physiotherapy & frequently asked questions |



ABOUT YOUR SURGEON

Dr. Harshil Shah

Consulting Orthopaedic Surgeon specialising in joint replacement, arthroscopy and trauma. This guide reflects the conventional, anterior-approach, robotic-assisted and day-care techniques used in his Ahmedabad practice — so you can walk into your consultation already informed.



M.S. Orthopaedics

Postgraduate specialisation in orthopaedic surgery.



FIJR

Fellowship in Joint Replacement.



FIAS

Fellowship in Arthroscopic Surgery.

“Use the questions this guide raises to make the most of your time with your surgeon — every patient’s anatomy, health and recovery pace are different.”

GETTING STARTED

Do you need a hip replacement?

Hip replacement is performed mainly to relieve pain and stiffness caused by hip arthritis, and is sometimes used to treat fractures or other degenerative conditions.

“When the cartilage lining the ball and socket wears down, the resulting osteoarthritis can make it painful simply to walk or to get in and out of a chair.”



Consider it if you have

- ✓ Severe hip pain unrelieved by medication, interfering with work, sleep or daily life.
- ✓ Stiffness that restricts movement — walking, stairs, or putting on shoes and socks.



Before considering surgery

Surgery isn't always the first step. Anti-inflammatory medication and physical therapy provide meaningful relief for many patients — surgery becomes worth discussing once these measures stop controlling symptoms.



In the joint

Articular cartilage wears away and exposed bone surfaces rub together — the direct source of arthritic pain and stiffness.



In daily life

Walking, climbing stairs, sleeping comfortably and simple tasks become progressively harder as the joint deteriorates.

WHAT THIS GUIDE HELPS YOU DO



Recognise when symptoms are worth a surgical opinion.



Understand the approaches and implant options available.

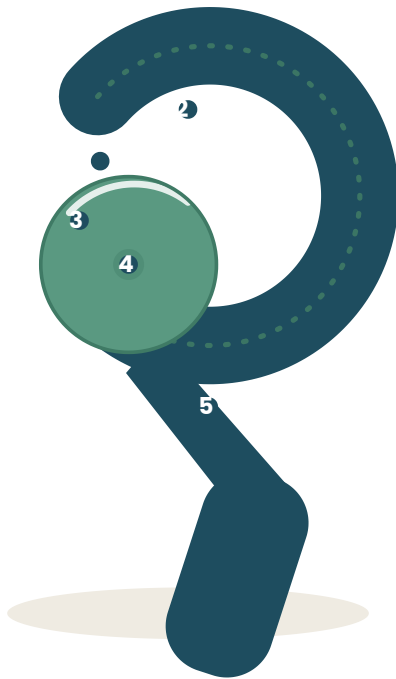


Plan realistically for surgery day and the weeks after.

THE BASICS

Understanding your hip

The hip is a ball-and-socket joint — one of the body's largest and most stable — designed to bear your body weight while allowing a wide range of movement.



1 Femoral head

The rounded top of the femur (thigh bone) — the "ball" of the joint.

2 Acetabulum

A cup-shaped part of the pelvis into which the femoral head fits — the "socket."

3 Articular cartilage

A smooth coating letting ball and socket glide together — worn away in osteoarthritis.

4 Labrum

A ring of cartilage deepening the socket rim, helping hold the ball securely in place.

5 Ligaments & capsule

A fibrous capsule holding the ball in the socket while allowing it to rotate freely.

! Clinical note — osteoarthritis wears the articular cartilage down until bone meets bone. This is the source of arthritic hip pain.

CHOOSING AN OPTION

Types of hip replacement

Four approaches to replacing or resurfacing the joint, chosen according to your anatomy, age, activity level and the extent of joint damage.



MOST COMMON

Total hip replacement

Both the femoral head and acetabulum are resurfaced — a stem and ball on the femoral side, a shell and liner on the socket side.



SELECT PATIENTS

Hip resurfacing

The femoral head is reshaped and capped with smooth metal rather than removed, preserving more bone for select, typically younger, active patients.



FRACTURE SETTING

Partial replacement

Hemiarthroplasty — only the femoral head is replaced; the natural socket is left in place. Used mainly for specific hip fractures in older patients.



BILATERAL / REVISION

Both hips, or a redo

Bilateral — both hips in one operation or staged, for healthy patients under ~75.
Revision — a redo for a loosened implant; lifetime risk is under 5%.



Which type is right for you? — your surgeon recommends a type based on X-rays, bone quality, age and activity level. Total hip replacement suits the great majority of patients with arthritis affecting the whole joint.

CHOOSING AN OPTION

Anterior vs. posterior approach

Whichever implant is placed, the surgeon needs a pathway to reach the joint. These are the two most common approaches for total hip replacement.



TRADITIONAL

Posterior approach

- ✓ Incision on the side/back of the hip; excellent surgical visibility.
- ✓ Some muscle and tendon splitting is typically required.
- ✓ Hip precautions commonly advised in the early weeks.
- ✓ Long-standing track record across nearly all patient types.



MUSCLE-SPARING

Anterior approach

- ✓ Incision at the front of the thigh, working between muscle planes.
- ✓ Minimises muscle splitting and tendon release.
- ✓ Most patients don't need significant hip-movement precautions.
- ✓ Well suited to fluoroscopy and robotic guidance.

Dislocation & outcomes — risk is low; some studies report a somewhat higher rate than anterior. By ~3 months, outcomes are excellent and equivalent.

Dislocation & outcomes — risk is low; robotic-assisted anterior technique shows very low dislocation rates in registry data. Outcomes converge by ~3 months.

A note on choosing between the two — don't choose a surgeon or hospital purely on marketing claims that one approach is automatically "better." Published long-term data shows comparable outcomes for a well-performed approach either way. Ask directly which technique your surgeon recommends for your anatomy, and why.

MATERIALS & TECHNOLOGY

Implants & robotic assistance

A total hip implant has three components, and robotic guidance is increasingly used to help place them with precision.



FEMORAL SIDE

Stem & ball

Stem — inserted into the femur, usually titanium, cobalt-chrome or stainless steel; cemented in weaker bone, or grown onto naturally otherwise. Ball — sits atop the stem, usually cobalt-chrome or ceramic.

SOCKET SIDE

Shell & liner

Typically a titanium shell — onto which the pelvic bone grows — lined with a high-density polyethylene liner. Screws may be added through the shell for extra fixation where needed.

20 yrs

Modern ceramic-on-polyethylene implants show no significant wear even after 20 years in studies — most are expected to last a lifetime.



The role of robotic assistance

Robotic and computer-navigation systems give the surgeon real-time, 3-D guidance for positioning implant components precisely, based on your individual anatomy — while the surgeon remains in full control throughout. Robotic assistance has been linked to improved implant positioning, alignment and ligament balance, and, in a large registry study, a lower risk of needing revision surgery for dislocation compared with manual technique.

A note on choosing — robotic assistance is a precision tool, not a guarantee of a better outcome by itself. Outcomes after hip replacement are excellent overall whether or not a robot is used.

RECOVERY PATHWAY

Day-care hip replacement — ERAS

Enhanced Recovery After Surgery (ERAS) is the evidence-based pathway behind modern day-care (same-day) hip replacement, allowing most healthy patients to safely return home the same day.

Traditional inpatient	Day-care / ERAS
Hospital stay	One to several days
Pain control	Leans more on opioid medication
Mobilisation	Typically begins the next day
Environment	Longer hospital exposure
Discharge	Same day, once checkpoints are met
Pain control	Multimodal, opioid-sparing — often 4/10 or under
Mobilisation	Most walk with a cane the same day
Infection risk	Lower, with a shorter stay

ARE YOU A CANDIDATE?

- ✓ Prefer initial recovery at home rather than in hospital.
- ✓ Have family or a friend to support you at home.
- ✓ Generally healthy, without significant uncontrolled conditions.
- ✓ Your surgical & anaesthesia team confirms eligibility individually.

✓ **Not automatically right for everyone** — you're only discharged once you clear defined checkpoints: pain is well-controlled, you can transfer safely, and manage stairs if needed. Staying an extra night, if a checkpoint isn't met, is a normal, safe part of the pathway — not a failure of the plan.

THE FULL PICTURE

Your journey — pre-op to post-op

A structured, checkpoint-based path from your first consultation through to your six-week follow-up and beyond.



BEFORE SURGERY

Consultation

Clinical exam and imaging confirm the diagnosis. Non-surgical options are discussed first; if surgery is right for you, approach, robotic assistance and day-care eligibility come next.



1-3 WEEKS BEFORE

Pre-operative optimisation

Blood tests and anaesthesia fitness check; counselling on safe transfers; home planning for support and equipment.



SURGERY DAY

Day of surgery

Arrival, final checks and multimodal anaesthesia planning. Surgery typically takes about 1.5 hours (~2 hours including setup).



0-24 HOURS POST-OP

First hours after surgery

Opioid-sparing pain control begins immediately. Early mobilisation — most walk with a cane the same day or by the next morning — before safe transfer and stair technique is taught.



SAME DAY, OR SHORT STAY

Discharge

Clear written instructions on wound care, medication and warning signs. A follow-up and physiotherapy plan is scheduled before you leave.



WEEKS 1-6

Early recovery at home

Daily physiotherapy continues; cane use typically needed 2-3 weeks. Driving generally resumes at 2-4 weeks. Follow-up with your surgeon around six weeks.



6 WEEKS - 3 MONTHS

Ongoing recovery

Strength, walking distance and confidence keep building. Return to active sport and more demanding activity generally possible by around three months.

GETTING YOU MOVING

Physiotherapy directions

Recovery pace varies from patient to patient — always follow the specific exercise plan, precautions and timelines given by your surgeon and physiotherapist.

**BEFORE SURGERY****Pre-habilitation**

Gentle strengthening for the hip and thigh, walking or stationary cycling for stamina, and practising the safe transfer techniques you'll use right after surgery.

**DAY 0-14****Immediately after surgery**

Guided walking with a cane, walker or crutches, increasing distance daily. Ankle pumps and gentle hip and knee movement aid circulation and ease stiffness.

**WEEKS 2-6****Building strength**

Progressive strengthening for the hip, thigh and core; supported stair practice and balance work, with walking aids gradually reduced.

**BEYOND 6 WEEKS****Returning to activity**

Continued strengthening with low-impact activity — walking, swimming, cycling — and return to more demanding activity or sport from around three months.



Follow your team's precautions — even once you feel comfortable. Most re-injuries happen when patients push beyond what's been advised too early.

BEFORE YOUR CONSULTATION

Frequently asked questions

The questions patients ask most often.

What does the surgery involve?

The damaged ball (top of the femur) and socket (part of the pelvis) are removed and replaced with prosthetic implants, relieving pain and restoring mobility.

How do I know if I need one?

Consider it for severe pain medication doesn't relieve, or stiffness limiting daily activity. Non-surgical options are worth trying first.

Is the anterior approach really better?

Its muscle-sparing nature supports fewer restrictions and quicker early gains. Outcomes converge by ~3 months — no approach is proven superior for everyone.

Will I need "hip precautions"?

Most anterior-approach patients do not. Posterior-approach patients are more commonly advised — your team will confirm what applies to you.

Does the robot perform the surgery?

No. It's a guidance tool helping the surgeon plan and verify implant position with sub-millimetre precision — your surgeon stays in full control.

How long do implants last?

Modern ceramic-on-polyethylene implants show no significant wear even after 20 years, and most are expected to last a lifetime.

What is day-care hip replacement?

Going home the same day as surgery — lower risk of hospital-acquired infection and a quicker, more comfortable return to normal activity.

How much pain will I have?

Modern multimodal pain management typically keeps reported pain to no more than about 4/10 on average, with staff on hand to address discomfort promptly.

How soon can I drive again?

On average 2–4 weeks — once you're off narcotic pain medication and able to react quickly, often sooner after a left-hip replacement.

How successful is the surgery overall?

Very — about 95% of patients experience significant pain relief, and over 97% report improved mobility and quality of life two years on.



Still have questions? — bring this guide to your consultation. There's no such thing as too small a question when it's about your own hip.



Questions before surgery? Bring this guide to your consultation.

Every patient's anatomy, health and recovery pace are different. Use the questions this guide raised to make the most of your time together — we'll walk through it, one step at a time.



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